

IWOAT SUMMER SCHOOL 2026: AN INTRODUCTION TO STABLE HOMOTOPY THEORY AND BRAVE NEW ALGEBRA

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ABSTRACT. The purpose of this summer school is to provide a rapid introduction to the foundations of modern stable homotopy theory. We will begin with a treatment of abstract homotopy theory, covering both model categories and $(\infty, 1)$ -categories. Using this base, we will introduce spectra and stable categories from a variety of perspectives. We will then talk about the symmetric monoidal structure on the category of spectra and ring and module objects, both from an operadic and point-set perspective. This will include natural geometric examples, notably Thom spectra. Finally, we will give a streamlined treatment of Bousfield localization. The course will roughly follow the outline of the book Stable categories and structured ring spectra. External references include Hovey's 'Model Categories', Riehl's 'Categorical Homotopy Theory', Elmendorf-Kriz-Mandell-May (EKMM), Mandell-May-Schwede-Shipley (MMSS), Blumberg-Gepner-Tabuada (BGT), Ravenel's 'Nilpotence and Periodicity', and Waldhausen's 'Algebraic K -theory of spaces' (Springer LNM 1126).

Prerequisites: Basic algebraic topology at the level of A Concise Course (including homology, CW-complexes, cofibrations and fibrations, elementary category theory).

Day 1: Abstract homotopy theory.

- (i) Overview of the class.
- (ii) Model categories. The homotopy category, calculus of fractions, model categories, derived functors (Quillen functors), homotopy colimits and limits.
References: Sections 2.2–2.5.
- (iii) Simplicial (model) categories. Simplicial mapping spaces. Quillen's SM7. Dwyer-Kan simplicial localization.
References: Sections 2.3.5, 2.4.4, 2.6. For SM7, see Section 4.2 of Hovey and Section III.11.4 of Riehl.
- (iv) $(\infty, 1)$ -categories. Quasicategories, the homotopy category, mapping spaces, homotopy limits and colimits, comparisons with model and simplicial categories.
References: Sections 2.7–2.10.

Day 2: Spectra and stable categories.

- (i) Diagram spectra; symmetric, orthogonal, excisive functors. The smash product of spectra. Symmetric monoidal model categories.
References: Sections 3.1, 3.5, 3.7–3.8, Section 4.1 for excisive functors,

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Section 3.3.2 for symmetric monoidal model categories. See also MMSS for the canonical source.

- (ii) Γ -spaces. Comparison to diagram spectra, the smash product of Γ -spaces. The model structure on Γ -spaces.
References: Sections 3.4, 3.9, 3.10. Supplement with MMSS.
- (iii) Stable model categories, SM7 revisited. Mapping spectra. Spectral categories. Categories of spectra in model categories.
References: Sections 3.2-3.3, 3.6.
- (iv) Spectra and excisive functors in $(\infty, 1)$ -categories. The symmetric monoidal structure. The universal property of spectra. Stabilization. Stable ∞ -categories. Triangulated categories and t -structures.
References: Sections 4.1-4.7.

Day 3: Rings and modules in stable homotopy theory, part one.

- (i) The model structures on rings and modules in diagram spectra. The positive model structure. Homotopy limits and colimits in rings and commutative rings. Constructions with rings and modules. Change-of-rings and derived categories.
References: Sections 6.1-6.3, Section 3.10.5. For homotopy limits and colimits in categories of (commutative) rings, see Section VII.2 of EKMM.
- (ii) Morita theory, module categories, bar constructions and cyclic bar constructions. Spectral categories vs. stable ∞ -categories.
References: Sections 1, 2 and 4 of BGT.
- (iii) Operads. Operadic algebras and modules. Model structures on operadic algebras. Rectification. Operadic spaces and Γ -spaces.
References: Sections 5.1-5.11; Sections 3.10.2 for Γ -spaces.
- (iv) Eilenberg-Mac Lane spectra, HM -modules, and chain complexes. Dold-Kan. Algebraic vs. non-algebraic stable categories.
References: Section 6.3; Section 2.4 for the Dold-Kan.

Day 4: Rings and modules in stable homotopy theory, part two.

- (i) Units and Thom spectra. The unit space, spectrum. The defining adjunction. The Thom spectrum as a symmetric monoidal functor.
References: Section 6.4.
- (ii) THH and TAQ ; construction, definition, basic properties. THH as the tensor for commutative rings. Examples of TAQ .
References: Sections 6.6-6.8.
- (iii) Algebraic K -theory; inputs as Waldhausen categories, stable categories. The structural theorems and the universal property. The cyclotomic trace.
References: Section 6.6 for the trace from algebraic K -theory to THH /cyclic-bar context; Springer LNM 1126 for Waldhausen K -theory and the basic theorems, BGT for the universal property and the cyclotomic trace.
- (iv) Picard and Brauer groups, generalized Thom spectra. Examples and computations; the E_∞ -ring structure on ku via bisymmetric monoidal categories.
References: Sections 6.4, 6.5, and 6.9.

Day 5: Bousfield localization and further directions.

- (i) Localization and local objects. Model structure approach via the small object argument. Examples; p -local and p -complete spectra. The rational homotopy category and Sullivan's theorem.
References: Sections 7.1–7.7; some examples in Sections 7.9–7.10; Section 5.12.
- (ii) Localization in the stable setting, stable ∞ -categories and localization, symmetric monoidal structures.
References: Sections 7.8–7.12.
- (iii) A brief introduction to chromatic localizations and the chromatic perspective on the stable category.
References: Sections 7.1 and 7.9; Ravenel.
- (iv) Visions of spectral algebraic geometry.
References: Chapter 8.

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